



14. The sending end voltage in terms of receiving end parameters and generalized constants A, B, C, D is written as
 A) $V_s = A \cdot V_r + B \cdot I_r$ B) $V_s = A \cdot V_r - B \cdot I_r$
 C) $V_s = C \cdot V_r + B \cdot I_r$ D) $V_s = C \cdot V_r - D \cdot I_r$
15. If the conductor size is increased, the corona effect is
 A) Decreased C) increased
 C) remaining same D) none of these
16. In short transmission lines, the effects of are neglected
 A) Resistance B) Inductance
 C) Capacitance D) Both inductance and capacitance
17. If the length of a cable is doubled, its capacitance is
 A) Halved B) Doubled
 C) Quadrupled D) Remains same
18. In capacitance grading of cables, we use a dielectric
 A) Composite B) Homogeneous
 C) Both A and B D) None of these
19. While designing a distributor the main consideration is
 A) Voltage drop along its length B) The current carrying capacity
 C) Both A and B D) None of these
20. The major part of investment on secondary distribution is made on
 A) Conductors B) Pole fittings
 C) Distribution transformers D) Insulators

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* | | | | | | | | | | | | | | |
|---|--------------|------|-------|-------|-------|-------|-------|-----------|----|----|----|----|----|----|--|--|--|--|
| 1. a) With neat sketch describe the schematic arrangement of thermal power plant. | 8 | L*2 | 1 | 1,2 | | | | | | | | | | | | | | |
| b) A generating station has the following daily load cycle: | | | | | | | | | | | | | | | | | | |
| <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <th style="width: 10%;">Time (Hours)</th> <th style="width: 10%;">0-6</th> <th style="width: 10%;">6-10</th> <th style="width: 10%;">10-12</th> <th style="width: 10%;">12-16</th> <th style="width: 10%;">16-20</th> <th style="width: 10%;">20-24</th> </tr> <tr> <td>Load (MW)</td> <td>40</td> <td>50</td> <td>60</td> <td>50</td> <td>70</td> <td>40</td> </tr> </table> | Time (Hours) | 0-6 | 6-10 | 10-12 | 12-16 | 16-20 | 20-24 | Load (MW) | 40 | 50 | 60 | 50 | 70 | 40 | | | | |
| Time (Hours) | 0-6 | 6-10 | 10-12 | 12-16 | 16-20 | 20-24 | | | | | | | | | | | | |
| Load (MW) | 40 | 50 | 60 | 50 | 70 | 40 | | | | | | | | | | | | |
| Draw the load curve and find (i) maximum demand (ii) units generated per day and (iii) average load. | 8 | L3 | 2 | 1,2 | | | | | | | | | | | | | | |
| 2. a) Explain the nuclear power plant with neat schematic diagram. | 8 | L2 | 1 | 1,2 | | | | | | | | | | | | | | |
| b) The load on an installation is 800 kW, 0.8 lagging p.f. which works for 3000 hours per annum. The tariff is Rs 100 per kVA plus 20 paise per kWh. If the power factor is improved to 0.9 lagging by means of loss-free capacitors costing Rs 60 per kVAR, calculate the annual saving effected. Allow 10% per annum for interest and depreciation on capacitors. | 8 | L3 | 2 | 1,2 | | | | | | | | | | | | | | |
| 3. a) Explain with the neat schematic diagram the functioning of hydroelectric power plant. | 8 | L2 | 1 | 1,2 | | | | | | | | | | | | | | |
| b) Enumerate the advantages of interconnected grid system. | 4 | L1 | 2 | 1,2 | | | | | | | | | | | | | | |
| c) A diesel station supplies the following loads to various consumers :
Industrial consumer = 1500 kW ;
Commercial establishment = 750 kW
Domestic power = 100 kW;
Domestic light = 450 kW
If the maximum demand on the station is 2500 kW and the number of kWh generated per year is 45×10^5 , determine (i) the diversity factor and (ii) annual load factor. | 4 | L3 | 2 | 1,2 | | | | | | | | | | | | | | |

Unit – II

- | | | | | | |
|-------|--|---|----|---|-----|
| 4. a) | Derive the expression for sag when conductors are suspended on supports at unequal levels. | 8 | L3 | 3 | 1,2 |
| b) | Define Corona discharge. Brief out the factors affecting Corona and methods to reduce corona effect. | 8 | L2 | 4 | 1,2 |
| 5. a) | In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self-capacitance of each insulator, find (i) the distribution of voltage over 3 insulators and (ii) string efficiency. | 8 | L3 | 3 | 1,2 |
| b) | Derive the expression for sending end voltage and current of a medium (T model) transmission line. | 8 | L2 | 4 | 1,2 |
| 6. a) | Derive the expression for inductance of a three-phase overhead line when conductors are symmetrically spaced. | 8 | L3 | 3 | 1,2 |
| b) | An overhead 3-phase transmission line delivers 5000 kW at 22 kV at 0.8 p.f. lagging. The resistance and reactance of each conductor is 4 Ω and 6 Ω respectively. Determine : (i) sending end voltage (ii) percentage regulation (iii) transmission efficiency. | 8 | L4 | 4 | 1,2 |

Unit – III

- | | | | | | |
|-------|--|---|----|---|-----|
| 7. a) | Derive the expression for dielectric stress in a single core cable | 8 | L3 | 5 | 1,2 |
| b) | A 2-wire d.c. distributor cable AB is 2 km long and supplies loads of 100A, 150A, 200A and 50A situated 500 m, 1000 m, 1600 m and 2000 m from the feeding point A. Each conductor has a resistance of 0.01 Ω per 1000 m. Calculate the p.d. at each load point if a p.d. of 300 V is maintained at point A. | 8 | L3 | 5 | 1,2 |
| 8. a) | State the classification of cables (according to voltage) and discuss commonly used insulating materials in underground cables. | 8 | L3 | 5 | 1,2 |
| b) | A single-core lead sheathed cable is graded by using three dielectrics of relative permittivity 5, 4 and 3 respectively. The conductor diameter is 2 cm and overall diameter is 8 cm. If the three dielectrics are worked at the same maximum stress of 40 kV/cm, find the safe working voltage of the cable.
What will be the value of safe working voltage for an ungraded cable, assuming the same conductor and overall diameter and the maximum dielectric stress? | 8 | L3 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
